Cross-Platform Emotion and Semantic Analysis of Social Media Comments

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Abstract

We create a dataset of comments taken from short-form media from various social media sites. We then classify them into a set of emotions using a pre-trained model, and perform analysis on the results.

1 Introduction

This project explores how users interact with short-form video content on TikTok, YouTube, Instagram, and Facebook. We investigate a central question: do user comments across these platforms share similar emotional and semantic characteristics, or does each platform foster a unique communication style?

We chose this topic because social media comments are short, informal, and often contain slang, emojis, irony, or ambiguous expressions. This makes them an interesting challenge for Computational Linguistics. In our opinion, a simple positive/negative/neutral sentiment analysis is not enough to describe these reactions, because users may express more specific emotions such as joy, anger, sadness, fear, surprise, or neutrality.

Our approach combines emotion classification, semantic representation, and manual evaluation. First, we collect and preprocess comments from multiple platforms, as described in Section 3.1. Then, we apply pretrained NLP models to extract emotions and sentence embeddings, as presented in Section 3.3.1. Finally, we compare the platforms using visual analysis and evaluate the predictions manually, as discussed in Section 3.3.3 and Section 3.3.4.

2 Related Work

The field has converged on several standard benchmarks, one of them being **GoEmotions** (Demszky et al., 2020), A large-scale dataset of 58,000 Reddit comments labeled with 27 emotion categories (+ neutral), developed by Google; but also **ISEAR**, **SemEval-2025 Task 11**, and **BRIGHTER**.

In recent years, transformer-based models have emerged as state-of-the-art in text analysis research. Models such as **BERT**, **RoBERTa** (Liu et al., 2019), and XLNet are pre-trained on large corpora in an unsupervised manner, and leverage contextual representations to model natural language.

DeBERTa-v3 achieves the highest accuracy of 87.1% and macro-F1 of 0.86 on GoEmotions, followed by RoBERTa with 86.7% accuracy and 0.85 macro-F1, and DistilBERT with 83.9% accuracy and 0.81 macro-F1, all outperforming the BERT baseline of 85.2% accuracy and 0.83 macro-F1.

Training a classifier on top of prompt-based encoders such as mE5 and BGE yields significantly better results than fully fine-tuning XLM-R and mBERT, particularly in multilingual settings. (Xue et al., 2025)

While large generative models like Mistral 7B show promise with zero-shot and few-shot prompting, masked language models (MLMs) currently outperform them in sentence-level emotion classification. (Creanga and Dinu, 2024)

3 Method

The core methodology of this project relies on a sequential Natural Language Processing (NLP) pipeline designed to extract, clean, and analyze

unstructured text data. The approach transitions from bulk data acquisition from a cloud database to processing using pre-trained transformer models and culminates in a comparative statistical analysis. Our process can be broken down in four main stages: data aggregation, text normalization, deep-learning feature extraction (semantic embeddings and emotion classification), aggregated comparative visualization. Moreover, a quality check for the accuracy of our models was done afterward in order to legitimize our findings. By prioritizing high-dimensional vector analysis and multi-class emotion tracking over basic keyword frequency, we captured the underlying "vibe" and semantic overlap between the four platforms.

Advantages of the Chosen Methodology:

- *Scalability and Modular Automation:* Utilizing a centralized cloud database enabled concurrent access, allowing the team to efficiently distribute tasks and process the 60,000-comment dataset in parallel. To optimize the workflow, the pipeline was divided into discrete, sequentially named scripts. This modular approach establishes a highly readable architecture, making it straightforward for third parties to understand, inherit, or extend the code. By decoupling extraction, preprocessing, and modeling, we could cache intermediate data states locally as .csv and .parquet files. This "run-once" structure bypassed redundant, time-consuming database queries and heavy transformer inference, while ensuring the preprocessing and training scripts remain flexible enough to ingest new or different datasets.

- *Semantic Depth over Lexical Matching:* By using sentence embeddings rather than traditional Bag-of-Words or TF-IDF methods, our similarity matrix accounts for context and meaning, successfully identifying when users are discussing the same concepts even if they use entirely different vocabulary.

- *Emotional Granularity:* Opting for a 28-class emotion model instead of a standard binary (positive/negative) sentiment analyzer provided the nuance necessary for social media. It allowed us to distinguish between closely related reactions, such as "amusement" versus "admiration," which define the distinct culture of each platform.

- *Data Preservation:* The preprocessing heuristic was specifically designed to demojize rather than delete emojis, preserving a critical layer of intent in short-form media communication. Similarly, measures were taken in order to short slang terms that might have been detected as other languages had they not been handled.

Disadvantages and Limitations:

- *Lack of Multimodal Context:* The most significant drawback is that the comments are analyzed in a vacuum. By stripping the text away from the actual video content, audio, and on-screen text, the models lose the contextual irony or sarcasm that is heavily prevalent on platforms like TikTok and Instagram.

- *Black-Box Interpretability:* Because we rely on deep-learning transformers, exact feature importance is difficult to extract. We can track what the model predicts, but mapping exactly which specific words triggered an "anger" classification is inherently obfuscated by the architecture.

- *Linguistic Bias:* Enforcing an English-only filter (with minor exceptions for slang) to maintain model accuracy inherently discards a portion of the raw data. This introduces a sampling bias, potentially skewing the representation of globally diverse platforms.

- *Asymmetric Data Retention:* While the initial extraction gathered an even 15,000 raw comments per platform, the aggressive preprocessing and language filtering resulted in unequal final datasets (ranging from roughly 8,900 to 12,000 comments). This discrepancy in sample size means the platforms are not perfectly balanced, which could introduce slight statistical weighting biases when calculating semantic centroids or evaluating emotion distributions.

3.1 Dataset

The dataset was made by scrapping comments from 4 social media sites, **YouTube**, **TikTok**, **Instagram**, and **Facebook**. More specifically, comments were taken from the short-form media served by these sites, such as **Shorts** or **Instagram Reels**. Approximately 60000 comments were collected,

15000 per site. For each site, each comment was also chosen from a selection of 5 categories: **gaming**, **sport**, **tech**, **education** and **sport**.

The scrapping was done either through an API, or using **playwright** to obtain the comments from the page itself.

A selection of roughly 1000 of the comments from the entire dataset was manually annotated, which was used to compare and analyze the performance of the model on it.

Considering the data, we expect elements such as emojis or common abbreviations to be relevant for the model.

3.2 Preprocessing

Due to the nature of the model and the way it was trained, being a transformer, minimal processing was done on the raw text. However, stuff that offered minimal semantic information was either enhanced or removed. Emojis were replaced with their text form using the demojize function from the **emoji python library**, and links and tags/mentions were removed.

Platform-specific information, present due to the way the comments were obtained from the sites, was also removed, such as HTML tags and escape codes for **Youtube**, and sticker tags for **TikTok**. Preprocessing was also done to remove extra white space or non-standard punctuation.

After doing said cleanup, using the **langdetect library**, we then checked the remaining text to be in English to not trip up future model sentiment analysis with unknown words that would otherwise be marked as neutral. In order to protect certain small words like "lol" or "lmao", we allowed one- or 2-word comments to bypass this language check to protect platform-specific slang that otherwise might have failed the language check.

3.3 Method

3.3.1 Model Selection and Mechanics

To perform a dual-layered analysis of the extracted comments, we utilized two distinct pre-trained transformer models. We did not modify the underlying weights of these models, but we heavily adapted the inference pipeline for our specific hardware and data structure.

1. Emotion Classification:

We used SamLowe/roberta-base-go_emotions, a RoBERTa-based model fine-tuned on the GoEmotions dataset.

- Why it was chosen: Standard sentiment analysis (positive/negative/neutral) is too reductionist for short-form video comments. This model outputs probabilities across 28 distinct emotional categories, allowing us to capture specific nuances (e.g., distinguishing between "amusement" and "admiration"). This big variety also give us more to work with when visualizing and mapping data.

- How it works: As a RoBERTa model, it relies on a bidirectional self-attention mechanism to process the entire sequence of words simultaneously, capturing deep contextual relationships. The model outputs raw logits, which we passed through a softmax function to extract the single highest-probability emotion and its corresponding confidence score for each comment.

2. Semantic Embeddings:

For text vectorization, we used all-mpnet-base-v2 from the SentenceTransformers library.

- Why it was chosen: MPNet frequently ranks at the top of SOTA benchmarks for semantic search and clustering. It provides an great balance of speed and high-quality semantic representations.

- It maps sentences into a 768-dimensional dense vector space. By converting text into mathematical vectors, sentences with similar meanings (even if using different vocabularies) are placed closer together in the vector space.

3.3.2 Hyperparameters and Hardware Adaptations

Given the scale of the dataset (approx. 60,000 rows) and hardware limitations where the training was run, standard CUDA/NVIDIA execution was not viable. We instead utilized torch_directml to force the computations onto a high end AMD GPU.

- Batch Size: Set to 64. This was a necessary hyperparameter choice to balance processing speed against the VRAM limitations of the local GPU.

- Sequence Length: Truncation and padding were applied with a max_length of 512 tokens, which is the architectural limit of the RoBERTa model.

- Memory Management: To prevent out-of-memory (OOM) crashes during the MPNet embedding loop, we implemented manual tensor movement and explicit Python garbage collection (gc.collect()) after every batch.

3.3.3 Experiments and Baselines

Our primary experiment involved quantifying the semantic overlap between the four platforms.

We calculated the "Semantic Centroid" for each platform by taking the mean across the 768 dimensions of all comment embeddings belonging to that platform. We then computed the *Cosine Similarity* between these centroids:

$$\text{similarity} = \cos(\theta) = \frac{\vec{A} \cdot \vec{B}}{\|\vec{A}\|\|\vec{B}\|}$$

Where $\vec{A}$ and $\vec{B}$ represent the centroid vectors of two different platforms. This generated a similarity matrix (see Figure 1), outputting scores between 0 (completely unrelated) and 1 (identical discourse).

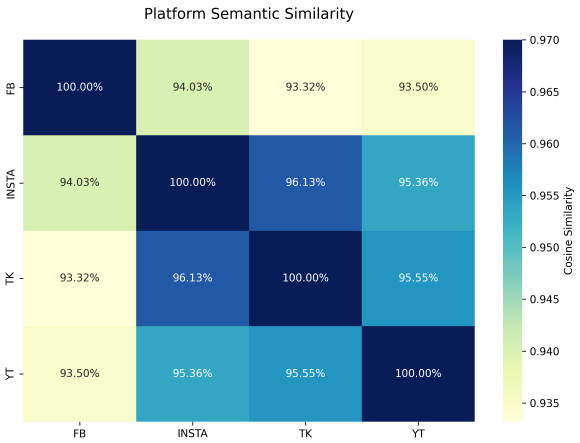

Figure 1: Cosine similarity matrix representing the semantic overlap of comment centroids across platforms.

Baseline Comparison: To ensure our emotion model wasn't blindly generating high-confidence inaccuracies (which would skew the distribution data), we did not rely solely on the model's theoretical SOTA accuracy. As detailed in section 3.1, our baseline was a manually annotated dataset of 1,000 comments (250 per platform). We mapped the model's 28 granular predictions into 6 macro "Pillars" (Joy, Anger, Sadness, Fear, Surprise, Neutral) and cross-referenced them against the human annotations. This ground-truth baseline provides a realistic metric of the model's actual performance on heavily slanged, short-form data.

On the annotated dataset, we obtained a macro-F1 score of 0.758, typical of scores reported for non-fine-tuned BERT-like models.

One should notice the accuracies and misclassifications obtained on the different emotion classes. Due to the nature of the dataset, which was obtained from popular short-form videos, the primary emotions present and identified in it are 'joy' and 'neutral', with 'fear' being nearly missing (2).

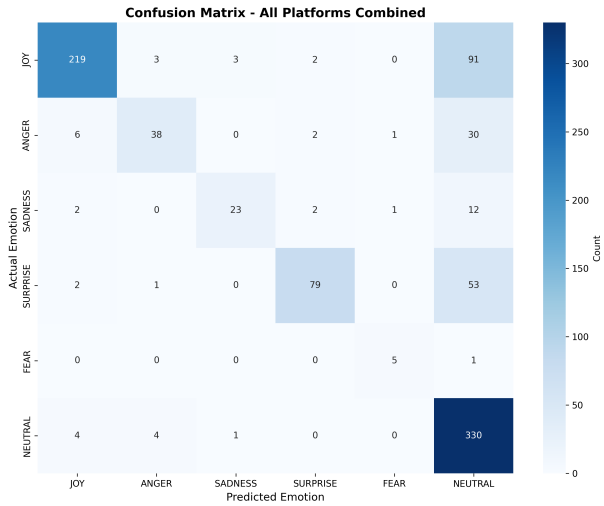

Figure 2: Confusion matrix for the classified emotions.

We noted that the comments in our dataset are in large amounts composed of ironic or sarcastic remarks and questions, or remarks that state opinions in an intentionally neutral way, even though a background emotion is expressed. This explains the misclassification of most of the emotions into the 'neutral' category.

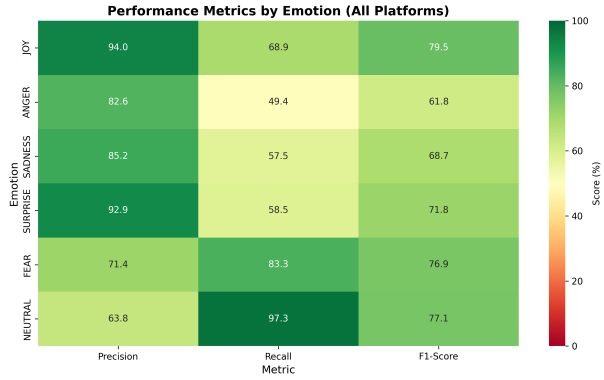

Figure 3: Precision and recall for the emotions.

One can observe that the model is biased toward the neutral classification, suggesting that it detects only the most obvious / strongly expressed emotional samples, and that subtle emotional expressions are being misclassified (3).

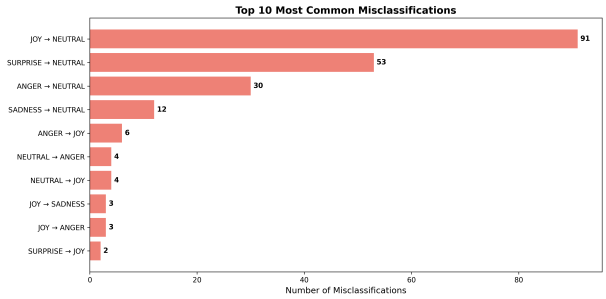

Figure 4: Top 10 emotion misclassifications.

The top, non-neutral, emotion misclassifications are 'anger->joy', 'joy->sadness', and 'joy->anger'; again, lining up with the presence of sarcastic and ironic text in the dataset (4).

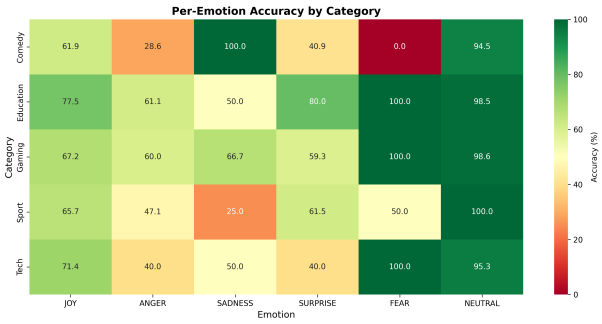

Figure 5: Emotion accuracy per category.

Finally, we also analyzed the behaviour per category (5). 'Sadness-comedy' had good accuracy, alongside 'joy-education' and 'surprise-education', probably because those emotions are usually expressed clearly and sincerely in those situations. The equivalent opposite argument can be made for 'anger-comedy' and 'sadness-sport', text in these categories being either sarcastic or intentionally unclear.

3.3.4 Feature Importance and Interpretability

Due to the architecture of deep-learning transformers, standard feature importance (i.e., mapping exactly which word contributed most to the output) is inherently "black-box." Without implementing complex gradient-tracking frameworks (like Captum), we cannot manually interpret individual word weights for a given classification.

To compensate for this, we interpret the data on a macro scale using Platform Emotion Distribution. By calculating the frequency percentage of each primary emotion per platform (Figure 3), we bypass the need to understand single-comment predictions and instead identify aggregate behavioral trends—such as one platform over-indexing in "amusement" while another over-indexes in "anger."

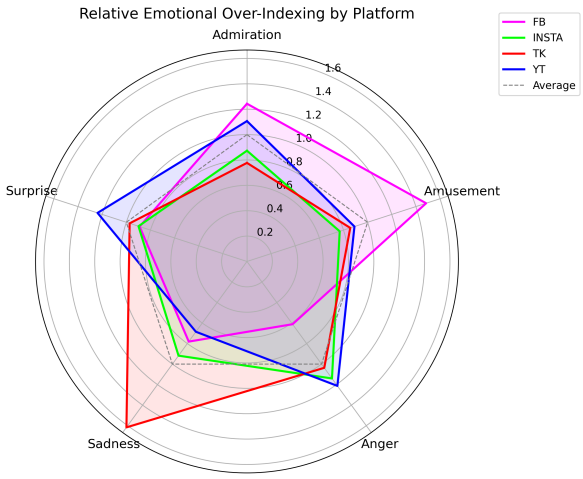

Figure 6: Relative emotional over-indexing, highlighting aggregate behavioral trends and primary emotions across platforms.

In conclusion, we cannot manually interpret the results at the individual comment level (we do not know exactly why a specific sentence was labeled "fear"). However, we can and do manually interpret the results at the macro level; the aggregated data provides a definitive, human-readable profile of each platform's overarching emotional culture, despite the model's internal mechanics remaining obfuscated.

4 Future Work

This work could be continued by expanding the dataset, making a larger subset of it annotated too. Including more emotion labels, such as all the ones included in the GoEmotions corpus, would also help make it a more robust dataset.

This dataset(or an expanded one mentioned above) could be used for the training of similar transformers to the one used in this project, offering a wider variety of training data, as GoEmotions contains only comments from Reddit.

5 Conclusion

This project served as a comprehensive exploration of the lifecycle of data, from the initial struggle

of acquisition to the finality of statistical visualization. Through the process of building this pipeline, several key insights were gained regarding both the technical mechanics of NLP and the practical realities of academic research.

What We Learned

Beyond the implementation of transformer architectures, the project provided a deeper understanding of the logic governing modern Natural Language Processing. Moving from a theoretical understanding to a practical application highlighted how models like RoBERTa leverage bidirectional context, evaluating words in relation to their neighbors rather than in isolation. We gained significant insight into how unstructured data is "tamed" through preprocessing, particularly the necessity of decoding emojis and handling platform-specific slang to maintain semantic integrity. Perhaps the most striking realization was the critical role of context; witnessing the model misclassify comments because it lacked the visual or auditory information of the video underscored the limitations of text-only analysis.

Successes and Interests

From a technical standpoint, the data gathering and preservation phase was the most engaging. Leveraging Playwright and OSINT-adjacent scraping methods to extract data from ephemeral, short-form media felt like a meaningful exercise in digital preservation. There is a specific satisfaction in capturing and normalizing volatile social media discourse into a structured, permanent dataset. Additionally, while the metrics were not revolutionary, the process of plotting "semantic centroids" and visualizing the emotional "vibe" of different platforms provided a tangible payoff for the engineering effort involved.

Challenges and Reflections

The project was not without its frustrations. The model training phase, while functional, felt constrained by the broader context of our academic environment. Managing this project alongside a Bachelor's Thesis and other final-year commitments meant that the "bandwidth" for deep architectural experimentation was limited. This resulted in a workflow that occasionally felt more like an exercise in pipeline management than creative model development.

Furthermore, the reality of working within a limited timeframe highlighted the difficulty of manual annotation and the inherent bias of a small, non-diverse team. While the final pipeline is modular and scalable, the experience served as a reminder that the most sophisticated models are still beholden to the quality of human-labeled ground truths and the contextual nuances they often overlook.

Limitations

Due to the small size of our team, only a limited amount of data could be annotated.

As mentioned above, some of the sites required the use of **playwright** to scrape them, which is a manual and unreliable method, since the layout of the site could change, not only through updates, but also through notifications or any disturbances that could happen during the scraping process.

There exist more professional scraping methods, and with a bigger team and a budget, it would be significantly easier to expand the dataset.

Due to the specific types of content we extracted the comments from, there exists a large bias on specific emotions, such as **Joy** and **Neutral**.

Ethical Statement

A potential unethical use of the dataset would be for training an LLM on it.

Due to the very small sample size of annotators, bias could come from each individual's opinions on each comment. To try to combat this, the annotations created by each member was double-checked by the others.

Additional bias was introduced by the fact that all of the members of this project are male, which could influence the way we classified what emotion each comment was.

Acknowledgements

We thank **The University of Bucharest**, and our course professor **Liviu Petrișor Dinu** and assistant **Zavelva-Miruna Andreea** for guiding us through this project.

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A Appendix

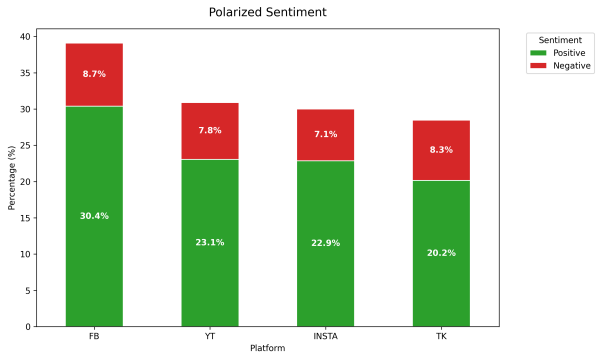
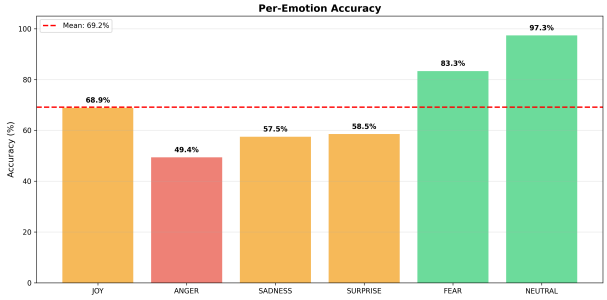
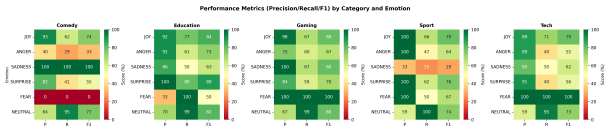
The repo for the project and dataset can be found at **https://github.com/laur004/Proiect-NLP**.

Here are additional tables and graphs that were obtained from analyzing the dataset with the model:

A.1 Per-Platform Per-Emotion Accuracy (%)

Platform	Joy	Anger	Sadness	Fear	Neutral	
YouTube	59.49	60.00	60.00	63.16	50.0	94.67
TikTok	71.43	55.56	80.00	60.00	100.0	97.27
Instagram	60.00	21.05	33.33	50.00	0.0	98.63
Facebook	83.87	58.62	70.00	58.82	100.0	98.77

A.2 Overall classification report (All Platforms)

Emotion	Precision	Recall	F1	Support
Joy	0.940	0.689	0.795	318
Anger	0.826	0.494	0.618	77
Sadness	0.852	0.575	0.687	40
Surprise	0.929	0.585	0.718	135
Fear	0.714	0.833	0.769	6
Neutral	0.638	0.973	0.771	339
Aggregate				
Accuracy			0.758	915
Macro avg	0.817	0.692	0.726	915
Weighted avg	0.812	0.758	0.755	915

A.3 Graphs